Digital Audio-Video Signal Processing

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1 Question - 12 pts

Load the *rice* image (hint: `I = imread('rice.png');`)

- Simulate a simplified JPEG image coding algorithm (encoding and decoding) that consists of the following steps:
  - Block-wise 2D-DCT transform (block size = 8).
  - Uniform scalar quantization. Step size $Q = 10$.
  - De-quantization.
  - Inverse 2D-DCT transform.
  - DPCM coding of DC coefficients.
- Changing the value of Q, plot a graph where Q is on the x-axis and $PSNR(dB)$ is on the Y axis.